

# Discretizing Problems Using Basis Functions

## Synopsis

Techniques for discretizing continuous inverse problems characterized by Fredholm integral equations of the first kind using continuous basis functions are discussed, both for general basis functions and for representers. The Gram matrix is defined and used in analyzing these problems. The method of Backus and Gilbert is also introduced.

## 5.1. DISCRETIZATION BY EXPANSION OF THE MODEL

To this point, we have discretized continuous inverse problems using the collocation scheme discussed in Chapter 1. In this chapter, we discuss an alternative approach in which we expand the unknown model  $m(x)$  in terms of a set of basis functions.

We begin with a Fredholm integral equation of the first kind,

$$d(x) = \int_a^b g(x, \xi) m(\xi) d\xi. \quad (5.1)$$

Note from Section A.11 that we can generalize linear algebra, including the concepts of linear independence, dot products, orthogonality, and the 2-norm, to spaces defined by functions. We will assume that our model  $m(x)$  lies in the space  $L_2(a, b)$  of functions that are square integrable on  $[a, b]$ .

In  $L_2(a, b)$ , the dot product of two functions  $f(x)$  and  $g(x)$  is defined as

$$\langle f(x), g(x) \rangle = \int_a^b f(x)g(x) dx, \quad (5.2)$$

the 2-norm of  $f(x)$  is

$$\|f(x)\|_2 = \sqrt{\int_a^b f(x)^2 dx}, \quad (5.3)$$

and the distance between two functions under this norm is

$$\|f(x) - g(x)\|_2 = \sqrt{\int_a^b (f(x) - g(x))^2 dx}. \quad (5.4)$$

Ideally, we could try to solve the Tikhonov regularization problem

$$\min \|m(x)\|_2 \quad \left\| \int_a^b g(x, \xi) m(\xi) d\xi - d(x) \right\|_2 \leq \Delta. \quad (5.5)$$

However, in practical cases, we do not fully know  $d(x)$  in the sense that we normally have a finite number of data points  $d_1 = d(x_1)$ ,  $d_2 = d(x_2)$ ,  $\dots$ ,  $d_m = d(x_m)$  arranged in a vector  $\mathbf{d}$ . An appropriate approach is thus to minimize the model norm  $\|m(x)\|_2$  subject to a constraint on the misfit at the data points:

$$\begin{aligned} \min \|m(x)\|_2 \\ \|\mathbf{g}(m(x)) - \mathbf{d}\|_2 \leq \delta, \end{aligned} \quad (5.6)$$

where

$$\mathbf{g}(m) = \begin{bmatrix} \int_a^b g(x_1, \xi) m(\xi) d\xi \\ \int_a^b g(x_2, \xi) m(\xi) d\xi \\ \vdots \\ \int_a^b g(x_m, \xi) m(\xi) d\xi \end{bmatrix}. \quad (5.7)$$

Consider a finite dimensional subspace  $H$  of  $L_2(a, b)$ , with the basis

$$h_1(x), h_2(x), \dots, h_n(x) \quad (5.8)$$

so that any model  $m(x)$  in the subspace  $H$  can be written uniquely as

$$m(x) = \sum_{j=1}^n \beta_j h_j(x). \quad (5.9)$$

We will seek the model  $m(x)$  in  $H$  that has minimum norm and adequately fits the data.

Substituting (5.9) into (5.1), and considering only those points  $x_i$  where we have data, we obtain

$$d(x_i) = \int_a^b g(x_i, \xi) \left( \sum_{j=1}^n \beta_j h_j(\xi) \right) d\xi \quad i = 1, 2, \dots, m. \quad (5.10)$$

To simplify the notation, let

$$g_i(\xi) = g(x_i, \xi) \quad i = 1, 2, \dots, m \quad (5.11)$$

and

$$d_i = d(x_i) \quad i = 1, 2, \dots, m. \quad (5.12)$$

We interchange the integral and sum to get

$$d_i = \sum_{j=1}^n \beta_j \int_a^b g_i(x) h_j(x) dx \quad i = 1, 2, \dots, m. \quad (5.13)$$

If we let  $\mathbf{G}$  be the  $m \times n$  matrix with entries

$$G_{i,j} = \int_a^b g_i(x) h_j(x) dx, \quad (5.14)$$

then the discretized inverse problem can be written as

$$\mathbf{G}\boldsymbol{\beta} = \mathbf{d}, \quad (5.15)$$

which can be solved to find the functional expansion coefficients  $\boldsymbol{\beta}$  in (5.9) to construct the desired model.

Because of noise in the data, we do not normally desire, nor will we be generally able, to solve (5.15) exactly. Furthermore, we typically need to regularize our solution to the inverse problem. We could try to solve

$$\begin{aligned} \min \|\boldsymbol{\beta}\|_2 \\ \|\mathbf{G}\boldsymbol{\beta} - \mathbf{d}\|_2 \leq \delta. \end{aligned} \quad (5.16)$$

However,  $\|\boldsymbol{\beta}\|_2$  is generally not proportional to  $\|m(x)\|_2$ . Instead, note that

$$\begin{aligned} \|m(x)\|_2^2 &= \int_a^b m(x)^2 dx \\ &= \int_a^b \left( \sum_{i=1}^n \beta_i h_i(x) \right) \left( \sum_{j=1}^n \beta_j h_j(x) \right) dx \\ &= \sum_{i=1}^n \sum_{j=1}^n \beta_i \beta_j \int_a^b h_i(x) h_j(x) dx. \end{aligned} \quad (5.17)$$

If we let  $\mathbf{H}$  be the  $n$  by  $n$  matrix with

$$H_{i,j} = \int_a^b h_i(x) h_j(x) dx, \quad (5.18)$$

then

$$\|m(x)\|_2^2 = \boldsymbol{\beta}^T \mathbf{H} \boldsymbol{\beta}. \quad (5.19)$$

The matrix  $\mathbf{H}$  is called the **Gram matrix** of the functions  $h_i(x)$ . It is relatively easy to show that  $\mathbf{H}$  is a symmetric and positive definite matrix, and thus has a Cholesky factorization

$$\mathbf{H} = \mathbf{R}^T \mathbf{R}, \quad (5.20)$$

so that

$$\begin{aligned} \|m(x)\|_2^2 &= \boldsymbol{\beta}^T \mathbf{R}^T \mathbf{R} \boldsymbol{\beta} \\ &= \|\mathbf{R} \boldsymbol{\beta}\|_2^2. \end{aligned} \quad (5.21)$$

We can write the Tikhonov regularization problem as

$$\begin{aligned} \min \|\mathbf{R} \boldsymbol{\beta}\|_2 \\ \|\mathbf{G} \boldsymbol{\beta} - \mathbf{d}\|_2 \leq \delta. \end{aligned} \quad (5.22)$$

We can use the Lagrange multiplier technique (Appendix C) to turn this into an unconstrained minimization problem

$$\min \|\mathbf{G} \boldsymbol{\beta} - \mathbf{d}\|_2^2 + \alpha^2 \|\mathbf{R} \boldsymbol{\beta}\|_2^2, \quad (5.23)$$

where the regularization parameter  $\alpha$  is selected so that we obtain a solution that adequately fits the data. This problem can easily be solved using the Cholesky factorization of the normal equations, the generalized singular value decomposition, or iterative methods discussed in Chapter 6.

Note that if we choose an orthonormal basis of functions, then

$$H_{i,j} = \int_a^b h_i(x) h_j(x) dx = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases} \quad (5.24)$$

or  $\mathbf{H} = \mathbf{R} = \mathbf{I}$ , and the minimization problem simplifies to (5.16).

To minimize a model derivative norm (e.g.,  $\|m''(x)\|_2$  instead of  $\|m(x)\|_2$ ), note that because

$$m''(x) = \sum_{j=1}^n \beta_j h_j''(x), \quad (5.25)$$

we can simply use the second derivative Gram matrix

$$H_{i,j} = \int_a^b h_i''(x) h_j''(x) dx \quad (5.26)$$

instead of (5.18).

The simple collocation scheme described in Chapter 1 can also be thought of as a special case of this approach, in which the  $h_i(x)$  basis functions are “boxcar” functions centered at the measurement points  $x_i$ . A variety of other bases have been used in practice. For example, sine and cosine functions can be used to produce solutions that are Fourier series, and, for problems involving spherical symmetries, the spherical harmonic functions are often used. In recent years, many researchers have investigated the use of wavelet bases in the solution of inverse problems.

The choice of the finite dimensional subspace  $H$  and its basis is critical to the success of this method in solving practical problems. As we have seen, the method finds the model  $m(x)$  in the subspace  $H$  that minimizes  $\|m(x)\|_2$  subject to fitting the data. There is no guarantee that the model we obtain will actually have the minimum norm over the entire space  $L_2(a, b)$ . However, if the subspace includes a sufficiently large class of functions, the resulting solution may be an adequate approximation to the solution of (5.6).

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## 5.2. USING REPRESENTERS AS BASIS FUNCTIONS

A particularly interesting choice of basis is to use the representer (1.33) themselves as the basis functions. This approach results in a solution that does effectively minimize  $\|m(x)\|_2$  over the entire space  $L_2(a, b)$ .

Consider the space  $G$  spanned by the representer

$$g_j(\xi) = g(x_j, \xi) \quad j = 1, 2, \dots, m. \quad (5.27)$$

Let  $G^\perp$  be the perpendicular complement of  $G$ .  $G^\perp$  consists of the functions  $f(x)$  in  $L_2(a, b)$  such that

$$\int_a^b f(x)g_j(x) dx = 0 \quad j = 1, 2, \dots, m. \quad (5.28)$$

Thus every function  $f(x)$  in  $G^\perp$  is in the null space of the forward operator. Every function  $m(x)$  in  $L_2(a, b)$  can be written uniquely as

$$m(x) = m_1(x) + m_2(x) \quad (5.29)$$

where  $m_1(x)$  is in  $G$  and  $m_2(x)$  is in  $G^\perp$ . By the Pythagorean theorem,

$$\|m(x)\|_2^2 = \|m_1(x)\|_2^2 + \|m_2(x)\|_2^2. \quad (5.30)$$

Since any nonzero function  $m_2(x)$  would have no effect on the fit to the data but would increase  $\|m(x)\|_2$ , the optimal norm-minimizing solution to (5.6) will always have  $m_2(x) = 0$ . This means that minimizing  $\|m(x)\|_2$  over our subspace  $G$  will be equivalent to minimizing  $\|m(x)\|_2$  over all of  $L_2(a, b)$ .

The Gram matrix of the representer,  $\mathbf{\Gamma}$ , has entries

$$\Gamma_{ij} = \int_a^b g_i(x)g_j(x) \, dx. \quad (5.31)$$

In terms of this matrix,

$$\|m(x)\|_2^2 = \boldsymbol{\beta}^T \mathbf{\Gamma} \boldsymbol{\beta}, \quad (5.32)$$

so that the discretized Tikhonov regularization problem (5.22) becomes

$$\begin{aligned} \min \quad & \|\mathbf{R}\boldsymbol{\beta}\|_2 \\ & \|\mathbf{\Gamma}\boldsymbol{\beta} - \mathbf{d}\|_2 \leq \delta, \end{aligned} \quad (5.33)$$

where  $\mathbf{R}$  is the Cholesky factorization of  $\mathbf{\Gamma}$ .

Can we extend this approach to minimizing a derivative-based model roughness measure (e.g.,  $\|m''(x)\|$  as we did in (5.26)? It is still true that adding a nonzero function  $m_2(x)$  from  $G^\perp$  to  $m(x)$  will have no effect on the fit to the data. Unfortunately, derivatives of models in  $G$  and  $G^\perp$  will no longer generally satisfy the orthogonality relationship (5.30), that is,

$$\|m''(x)\|_2^2 \neq \|m_1''(x)\|_2^2 + \|m_2''(x)\|_2^2, \quad (5.34)$$

because orthogonality of two functions  $f(x)$  and  $g(x)$  does not imply orthogonality of  $f''(x)$  and  $g''(x)$ . Because adding a nonzero function  $m_2(x)$  from  $G^\perp$  might actually decrease  $\|m''(x)\|_2$ , minimizing  $\|m''(x)\|_2$  over our subspace  $G$  in this case will not be equivalent to minimizing over  $L_2(a, b)$ .

### 5.3. THE METHOD OF BACKUS AND GILBERT

The method of Backus and Gilbert [2, 126] is applicable to continuous linear inverse problems of the form

$$d(x) = \int_a^b g(x, \xi)m(\xi) \, d\xi, \quad (5.35)$$

where we have observations at points  $x_1, x_2, \dots, x_n$ . Let

$$d_j = d(x_j) \quad j = 1, 2, \dots, m \quad (5.36)$$

and

$$g_j(\xi) = g(x_j, \xi) \quad j = 1, 2, \dots, m. \quad (5.37)$$

Using (5.35), we can write the  $d_j$  in terms of the true model,  $m_{\text{true}}$ , as

$$\begin{aligned} d_j &= \int_a^b g(x_j, \xi) m_{\text{true}}(\xi) d\xi \\ &= \int_a^b g_j(\xi) m_{\text{true}}(\xi) d\xi. \end{aligned} \quad (5.38)$$

The Backus and Gilbert method estimates  $m_{\text{true}}(x)$  at some point  $\hat{x}$ , given the  $m$  data values  $d_j$ . The method produces model estimates that are linear combinations of the data elements

$$\hat{m} = \sum_{j=1}^m c_j d_j \approx m_{\text{true}}(\hat{x}) \quad (5.39)$$

where the coefficients  $c_j$  are to be determined.

Combining (5.37) and (5.38) gives

$$\begin{aligned} \hat{m} &= \sum_{j=1}^m c_j \int_a^b g_j(x) m_{\text{true}}(x) dx \\ &= \int_a^b \left( \sum_{j=1}^m c_j g_j(x) \right) m_{\text{true}}(x) dx \\ &= \int_a^b A(x) m_{\text{true}}(x) dx, \end{aligned} \quad (5.40)$$

where

$$A(x) = \sum_{j=1}^m c_j g_j(x). \quad (5.41)$$

The function  $A(x)$  is called an **averaging kernel**. Ideally, we would like the averaging kernel to closely approximate a delta function

$$A(x) = \delta(x - \hat{x}) \quad (5.42)$$

because, assuming exact data, (5.40) would then produce exact agreement between the true and estimated model ( $\hat{m} = m_{\text{true}}(\hat{x})$ ). Since this is not possible, we will instead select the coefficients so that the area under the averaging kernel is one, and so that the width of the averaging kernel around  $\hat{x}$  is as small as possible.

In terms of the coefficients  $\mathbf{c}$ , the unit area constraint can be written as

$$\int_a^b A(x) dx = \int_a^b \sum_{j=1}^m c_j g_j(x) dx = \sum_{j=1}^m c_j \left( \int_a^b g_j(x) dx \right) = 1. \quad (5.43)$$

Letting

$$q_j = \int_a^b g_j(x) dx, \quad (5.44)$$

the unit area constraint (5.43) can be written as

$$\mathbf{q}^T \mathbf{c} = 1. \quad (5.45)$$

Averaging kernel widths can be usefully characterized in a variety of ways [126]. The most commonly used measure is the second moment of  $A(x)$  about  $\hat{x}$ :

$$w = \int_a^b A(x)^2 (x - \hat{x})^2 dx. \quad (5.46)$$

In terms of the coefficients  $\mathbf{c}$ , this can be written using (5.43) as

$$w = \mathbf{c}^T \mathbf{H} \mathbf{c} \quad (5.47)$$

where

$$H_{j,k} = \int_a^b g_j(x) g_k(x) (x - \hat{x})^2 dx. \quad (5.48)$$

Now, the problem of finding the optimal coefficients can be written as

$$\begin{aligned} \min \quad & \mathbf{c}^T \mathbf{H} \mathbf{c} \\ \text{subject to} \quad & \mathbf{c}^T \mathbf{q} = 1, \end{aligned} \quad (5.49)$$

which can be solved using the Lagrange multiplier technique.

In practice, the data may be noisy, and the solution may thus be unstable. For measurements with independent errors, the standard deviation of the estimate is given by

$$\text{Var}(\hat{m}) = \sum_{j=1}^m c_j^2 \sigma_j^2, \quad (5.50)$$

where  $\sigma_j$  is the standard deviation of the  $j$ th observation.

The solution can be stabilized by adding a constraint on the variance to (5.49)

$$\begin{aligned} \min \quad & \mathbf{c}^T \mathbf{H} \mathbf{c} \\ & \mathbf{q}^T \mathbf{c} = 1 \\ & \sum_{j=1}^n c_j^2 \sigma_j^2 \leq \delta. \end{aligned} \quad (5.51)$$

Again, this problem can be solved by the method of Lagrange multipliers. Smaller values of  $\delta$  decrease the variance of the estimate, but restrict the choice of coefficients so that the width of the averaging kernel increases. There is thus a trade-off between stability of the solution and the width of the averaging kernel.

The method of Backus and Gilbert produces a model estimate at a particular point  $\hat{\mathbf{a}}$ . It is possible to use the method to compute estimates on a grid of points  $x_1, x_2, \dots, x_n$ . However, since the averaging kernels may not be well localized around their associated grid points, and may vary and overlap in complicated ways, this is not equivalent to the simple collocation method of model representation introduced in Chapter 1. Furthermore, such an approach requires the computationally intensive solution of (5.51) for each point. For these reasons, the method does not receive wide use.

### Example 5.1

For a spherically symmetric Earth model, the mass  $M_e$  and moment of inertia  $I_e$  are determined by the radial density  $\rho(r)$ , where

$$M_e = \int_0^{R_e} (4\pi r^2) \rho(r) dr \quad (5.52)$$

and

$$I_e = \int_0^{R_e} \left( \frac{8}{3} \pi r^4 \right) \rho(r) dr. \quad (5.53)$$

Using  $R_e = 6.3708 \times 10^6$  m as the radius of a spherical Earth, and supposing that from astronomical measurements we can infer that  $M_e = 5.973 \pm 0.0005 \times 10^{24}$  kg and  $I_e = 8.02 \pm 0.005 \times 10^{37}$  kg·m<sup>2</sup>, we will estimate the density of the Earth in the lower mantle (e.g., at  $r = 5000$  km), and core (e.g., at  $r = 1000$  km).

Equations (5.52) and (5.53) include terms that span an enormous numerical range. Scaling so that

$$\hat{r} = r/R_e \quad \hat{\rho} = \rho/1000 \quad \hat{I}_e = I_e/10^{37} \quad \hat{M}_e = M_e/10^{24}$$



gives

$$\hat{M}_e = 0.2586 \int_0^1 (4\pi \hat{r}^2) \hat{\rho}(\hat{r}) d\hat{r} \quad (5.54)$$

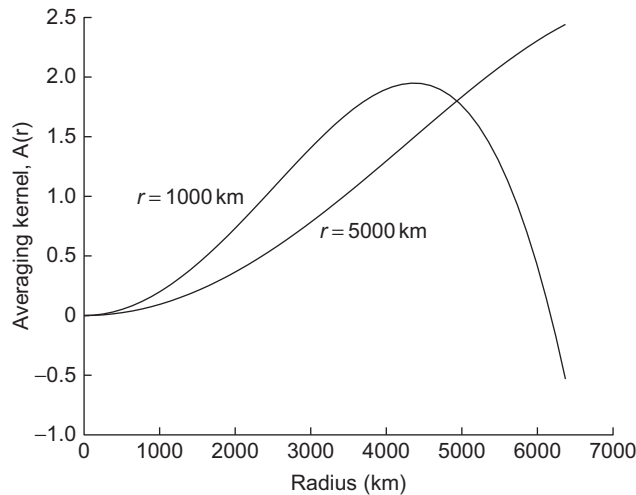
and

$$\hat{I}_e = 1.0492 \int_0^1 \left( \frac{8}{3} \pi \hat{r}^4 \right) \hat{\rho}(\hat{r}) d\hat{r}. \quad (5.55)$$

Applying (5.49) for  $r = 5000$  km gives the coefficient values  $\mathbf{c}^T = [1.1809, -0.1588]$  and a corresponding model density of  $5.8 \text{ g/cm}^3$ . This is a fairly accurate estimate of density for this radius, where standard earth models estimated using seismological methods [96] infer densities of approximately  $5 \text{ g/cm}^3$ . The associated standard deviation (5.50) is  $0.001 \text{ g/cm}^3$ , so there is very little sensitivity to data uncertainty.

At  $r = 1000$  km, we obtain the coefficients  $\mathbf{c}^T = [2.5537, -1.0047]$ , and a corresponding density estimate of  $7.2 \text{ g/cm}^3$ . This is not an accurate estimate for the density of the inner core, where standard earth models have densities of around  $13 \text{ g/cm}^3$ . The corresponding standard deviation is just  $0.005 \text{ g/cm}^3$ , so this inaccuracy is not related to instability in the inverse problem.

Figure 5.1 shows the averaging kernels corresponding to these model element estimates, and explains both the successful mantle and failed core density estimates. In the mantle case, the averaging kernel has much of its area near the targeted radius of



**Figure 5.1** Averaging kernels for target radii of 1000 and 5000 km.

5000 km. In the core case, however, the averaging kernel has most of its area at much greater  $r$ , and little area near the target radius of 1000 km. The fundamental reason for this situation is that both the mass and moment of inertia are relatively insensitive to the density of the innermost earth.

## 5.4. EXERCISES

1. Consider the earth density estimation problem of [Example 5.1](#). Apply the generalized inverse to estimate the density as a function of radius, using the given values of mass and moment of inertia. Obtain a density model composed of 20 spherical shells of equal thickness, and compare your results to a standard model.
2. Use the Gram matrix technique to discretize the integral equation from [Example 5.1](#).
  - a. Solve the resulting linear system of equations, and plot the resulting model.
  - b. What was the condition number of  $\mathbf{\Gamma}$ ? What does this tell you about the accuracy of your solution?
3. Show that if the representer  $g_i(t)$  are linearly independent, then the Gram matrix  $\mathbf{\Gamma}$  is nonsingular.
4. Show that if the basis functions in (5.9) are orthonormal, then  $\|m(x)\|_2 = \|\beta\|_2$ . Hint: Expand  $\|m(x)\|_2^2$  using (5.3), and then simplify using the orthogonality of the basis functions.
5. Recall the polynomial regression problem from Exercise 2.5. Instead of using the polynomials  $1, x, \dots, x^{19}$ , we will use the basis of Legendre polynomials, which are orthogonal on the interval  $[-1, 1]$ . These polynomials are generated by the recurrence relation

$$p_{n+1}(x) = \frac{(2n+1)xp_n(x) - np_{n-1}(x)}{n+1}, \quad (5.56)$$

starting with

$$p_0(x) = 1 \quad (5.57)$$

and

$$p_1(x) = x. \quad (5.58)$$

The next two Legendre polynomials are  $p_2(x) = (3x^2 - 1)/2$  and  $p_3(x) = (5x^3 - 3x)/2$ . This recurrence relation can be used both to compute coefficients of the Legendre polynomials and to compute values of the polynomials for particular values of  $x$ .

Use the first 20 Legendre polynomials to fit a polynomial of degree 19 to the data from Exercise 2.5. Express your solution as a linear combination of the Legendre

- polynomials and also as a regular polynomial. How well-conditioned was this system of equations? Plot your solution and compare it with your solution to Exercise 2.5.
6. Construct specific examples of functions  $m_1(x)$  and  $m_2(x)$  in  $L_2(-1, 1)$ , such that  $m(x) = m_1(x) + m_2(x)$ ,  $m_1(x) \perp m_2(x)$ , but  $\|m''(x)\|_2^2 \neq \|m_1''(x)\|_2^2 + \|m_2''(x)\|_2^2$ .

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## 5.5. NOTES AND FURTHER READING

Techniques for discretizing integral equations are discussed in [45, 126, 166, 172]. A variety of basis functions have been used to discretize integral equations including sines and cosines, spherical harmonics, B-splines, and wavelets. In selecting the basis functions, it is important to select a basis that can reasonably represent likely models. The basis functions must be linearly independent, so that a function can be written in terms of the basis functions in exactly one way, and (5.9) is thus unique. As we have seen, the use of an orthonormal basis has the further advantage that  $\|\beta\|_2 = \|m(x)\|_2$ .

The selection of an appropriate basis for a particular problem is a fine art that requires detailed knowledge of the problem as well as of the behavior of the basis functions. Beware that a poorly selected basis may not adequately approximate the solution, resulting in an estimated model  $m(x)$  that is very wrong. The choice of basis can also have a very large effect on the condition number of the discretized problem, potentially making it very ill-conditioned.

An important theoretical question is whether the solutions to discretized versions of a continuous inverse problem with noise-free data will converge to a solution of the continuous inverse problem. Engl, Hanke, and Neubauer provide an explicit example showing that nonconvergence of a discretization scheme is possible [45]. They also provide conditions under which convergence is guaranteed. For Fredholm integral equations of the first kind, the Gram matrix approach using representers can be shown to be convergent [45].